

1924/2172

Activity : 3Analytical Problem Solving

① Gradient magnitude and Direction

A) Given,

$$G_x = 24, G_y = 18$$

Gradient magnitude:-

$$G = \sqrt{G_x^2 + G_y^2}$$

$$= \sqrt{(24)^2 + (18)^2} \Rightarrow \sqrt{576 + 324} = \sqrt{900}$$

$$\therefore G = 30$$

Gradient Direction

$$\theta = \tan^{-1}(G_y/G_x)$$

$$= \tan^{-1}\left[\frac{18}{24}\right]$$

$$\theta = 36.87^\circ$$

② Non-Maximum Suppression:

A) Given;

Pixel: P_1 P_2 P_3 P_4 P_5

Magnitude: 25 50 90 60 30

$\Rightarrow$ Gradient direction is horizontal, so each Pixel is compared with its left and right neighbors

$\Rightarrow P_2 = 50$:- Neighbours 25 and 90 $\rightarrow 50$ is smaller

$\Rightarrow$ suppressed

$\Rightarrow P_3 = 90$:- Neighbours 50 and 60 $\rightarrow 90$ is smaller $\rightarrow$ left

$\Rightarrow P_4 = 60$:- Neighbours 90 and 30 $\rightarrow 60$ is smaller suppressed

50,

$P_3 = 90$ is retained.

$\Rightarrow$ other Pixels are suppressed at the boundaries (after comparison)

③ Double Thresholding

Given

• High Threshold = 100

Low Threshold = 50

Pixel	magnitude	classification
P_1	135	Strong edge
P_2	90	Weak edge
P_3	45	Non-edge
P_4	120	Strong edge
P_5	70	Weak edge
P_6	20	Non-edge

∴ Strong: P_1, P_4 if magnitude ≥ 100

Weak: P_2, P_5 if magnitude 50 - 99

Non-edge: P_3, P_6 if magnitude < 50

④ Edge Tracking by Hysteresis

Given

• High Threshold = 100 ; Low Threshold = 50

Step - 1 classification:

$P_1 = 130 \rightarrow$ strong

$P_2 = 95 \rightarrow$ weak

$P_3 = 40 \rightarrow$ non-edge

$P_4 = 115 \rightarrow$ strong

$P_5 = 65 \rightarrow$ weak

$P_6 = 25 \rightarrow$ non-edge

$P_7 = 105 \rightarrow$ strong

Strong: P_1, P_4, P_7

Weak: P_2, P_5

Non-edge P_3, P_6

Step 2: Hysteresis

• P_2 is strong edge

$P_1 \rightarrow$ keep P_2

P_5 is strong edge

$\rightarrow$ Remove P_5

Final edge Pixel: P_1, P_2, P_4, P_7

High Circle Transformation

① Calculate radius for each combination.

Formula: $r = \sqrt{(x-x_c)^2 + (y-y_c)^2}$

Given:

$$P_1 = (5, 4), P_2 = (7, 6)$$

$$C_1 = (2, 4), C_2 = (4, 3)$$

P_1 with C_1

$$r = \sqrt{(5-2)^2 + (4-4)^2} \\ = \sqrt{9} = 3$$

P_1 with C_2

$$r = \sqrt{(5-4)^2 + (4-3)^2} = \sqrt{1+1} = \sqrt{2} = 1.41$$

P_2 with C_1 :

$$r = \sqrt{(7-2)^2 + (6-4)^2} = \sqrt{25+4} = \sqrt{29} = 5.39$$

P_2 with C_2 :

$$r = \sqrt{(7-4)^2 + (6-3)^2} = \sqrt{9+9} = \sqrt{18} = 4.24$$

$\therefore C_1$ gets maximum matching radius value for the given

~~Points~~ Points: $r=3$ and 5.39 .

② Given, $P(8, 6)$, $C(4, 3)$

A) Formula: $r = \sqrt{(x-x_c)^2 + (y-y_c)^2}$

$$r = \sqrt{(8-4)^2 + (6-3)^2} \\ = \sqrt{4^2 + 3^2} \\ = \sqrt{16+9} = \sqrt{25}$$

$$r = 5$$

③ Given: $C = (5, 4)$

A) Center, $C = (5, 4)$

Edge Point, $P = (8, 8)$

Formula: $r = \sqrt{(x - x_c)^2 + (y - y_c)^2}$

$$r = \sqrt{(8 - 5)^2 + (8 - 4)^2}$$

$$= \sqrt{2^2 + 4^2}$$

$$= \sqrt{9 + 16}$$

$$= \sqrt{25}$$

$$r = 5$$

$\therefore$ The radius of the circle is 5.